Database Lab : Database Backup Execution & Audit Log Setup

Duration : 30 Min.

Scenario

As a Database Reliability Engineer at NIIT, after a failed backup alert, you must execute a schema backup of PostgreSQL using `pg_dump`, store the backup artifact in the backup directory, and record a backup audit log entry in the database.

Input Details

The environment has been pre-configured with the following local database platform instances:

- **Database Host:** `localhost` (Ubuntu Linux VM)
- **PostgreSQL Service:** `localhost:5432` | DB: `postgres` | User: `postgres` | Password: `PostgresPassword123!` | CLI: `psql`
- **Backup Directory:** `/var/backups/db/` (or `~/backups/`)

Username & Naming Conventions

Your candidate prefix is: **`<prefix>`** (e.g. `student-exam123`).

Resource naming: - **Backup Target Directory:** `/var/backups/db/<prefix>_backup/` (or `~/backups/<prefix>_backup/`) - **Backup File:** `postgres_schema.sql` - **PostgreSQL Audit Table:** `<prefix>_backup_audit`

Task Objectives

Perform the following backup and audit steps:

1. Create Backup Target Directory

- Create directory `/var/backups/db/<prefix>_backup/` (or `~/backups/<prefix>_backup/` if permissions require).

2. Execute pg_dump Schema Backup

- Run `pg_dump` to generate a schema-only backup of database `postgres`: `bash PGPASSWORD='PostgresPassword123!' pg_dump -h localhost -U postgres -d postgres --schema-only -f /var/backups/db/<prefix>_backup/postgres_schema.sql`

3. Record Backup Audit in Database

- Connect via `psql` and create table `<prefix>_backup_audit` with columns:
- `backup_id SERIAL PRIMARY KEY`
- `backup_file VARCHAR(200) NOT NULL`
- `backup_type VARCHAR(30) NOT NULL`
- `status VARCHAR(20) NOT NULL`
- `logged_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP`
- Insert a record logging `backup_file = 'postgres_schema.sql'`, `backup_type = 'SCHEMA_ONLY'`, and `status = 'COMPLETED'`.

Verification

Once you have performed the tasks, run the verification script:

```
python run.py
```

The verification system checks the backup file on disk and inspects the database audit table.

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Backup Directory & Target Path Configuration	4 Marks
TC2	pg_dump Schema Backup File Generation on Disk	3 Marks
TC3	Database Backup Audit Table & Sign-Off Record	3 Marks

Total Score: 10 Marks

Important Notes

- Replace `<prefix>` with your candidate prefix.
- Ensure the output SQL file contains valid PostgreSQL DDL statements.